1.2.1 ▶ ▶ ▶ ▶ Extension of $F = ma$

This rather short section concerns the full extended Newton's second law. It adds another layer of nuance that is important to note while considering Newtonian Mechanics.

The full, most rigorous representation of Newton's second law is the following:

$$\boxed{\mathbf{F}_{Net,External} = M\mathbf{a}_{cm}}. \tag{1.21}$$

Where M is the total mass of the system and $\mathbf{a}_{cm}$ is the acceleration of the center of mass of the system. To derive this equation, we first start with a system of particles. If we look at a single particle with mass m_i in this system, writing out Newton's second law gives

$$\mathbf{F}_{i,ext} + \mathbf{F}_{i,int} = m_i\mathbf{a}_i \tag{1.22}$$

where $\mathbf{F}_{i,ext}$ and $\mathbf{F}_{i,int}$ are the forces acting on the particle external and internal to the system respectively. Summing over all particles gives

$$\sum_i \mathbf{F}_{i,ext} + \sum_i \mathbf{F}_{i,int} = \sum_i m_i\mathbf{a}_i$$

$$\mathbf{F}_{Net,Ext} = M\mathbf{a}_{cm}. \tag{1.23}$$

The first term summed over the system becomes the net external force. The second term is equal to zero because of Newton's third law. The right hand side of the equation becomes $M\mathbf{a}_{cm}$ due to the definition of the center of mass i.e. $M\mathbf{a}_{cm} = \sum_i m_i\mathbf{a}_i$.

This equation states that the net force external to any system (rigid or non-rigid) equals the total mass of the system multiplied by the acceleration of the center of mass of the system. A lot of the time, it will be simply written as $F = ma$, but it is important to remember the full form of the equation ★